

Practice Sheet 1.3: Basics of Python Programming

☐ HARD LEVEL — Without Conditionals (Exam-Ready)

1. Net Salary with Multiple Allowances and Deductions

Problem Statement:

An employee's net salary is calculated as:

Net Salary = Basic + HRA (20%) + DA (10%) – PF (12%)

Write a Python program to calculate the **net salary**.

Input Format:

Basic salary (float)

Output Format:

Print the net salary.

2. Average Speed of a Journey

Problem Statement:

A vehicle travels two distances at two different speeds.

Write a Python program to calculate the **average speed** of the journey.

Input Format:

Distance1, Speed1

Distance2, Speed2

Output Format:

Print the average speed.

☐ *Hint:*

Total distance / Total time

3. Time Conversion (Seconds to Hours, Minutes, Seconds)

Problem Statement:

Write a Python program to convert a given number of seconds into **hours, minutes, and remaining seconds**.

Input Format:

Total seconds (int)

Output Format:

Print hours, minutes, and seconds.

□ *Concepts:* Integer division, modulo

4. Calculate Final Amount with Compound Interest

Problem Statement:

Write a Python program to calculate the **final amount** using the compound interest formula:
$$A = P(1 + R/100)^T$$

Input Format:

Principal (float)

Rate (float)

Time (int)

Output Format:

Print the final amount.

5. Convert Total Marks into Percentage and CGPA

Problem Statement:

A student has marks in 5 subjects, each out of 100.

Write a Python program to calculate the **percentage** and **CGPA**.

$$CGPA = \text{Percentage} / 9.5$$

Input Format:

Five subject marks

Output Format:

Print percentage and CGPA.

6. Maximum and Minimum of Three Numbers (Without Conditionals)

Problem Statement:

Write a Python program to find the **maximum** and **minimum** of three numbers **without using any conditional statements**.

Input Format:

Three numbers

Output Format:

Print maximum and minimum values.

☐ *Hint:* Use built-in functions.

7. Distance Between Two Points

Problem Statement:

Write a Python program to calculate the **distance between two points** (x1, y1) and (x2, y2) using the distance formula.

Input Format:

x1, y1

x2, y2

Output Format:

Print the distance.

8. Break Amount into Currency Notes

Problem Statement:

Write a Python program to calculate the **number of 100-rupee notes** and **remaining amount** for a given amount.

Input Format:

Amount (int)

Output Format:

Print number of 100-rupee notes and remaining amount.

□ *Concepts:* Division and modulo

9. Calculate BMI (Body Mass Index)

Problem Statement:

Write a Python program to calculate the **Body Mass Index (BMI)** using the formula:
$$\text{BMI} = \text{Weight} / (\text{Height} \times \text{Height})$$

Input Format:

Weight in kg (float)

Height in meters (float)

Output Format:

Print the BMI value.

10. Swap Two Numbers Without Using a Third Variable

Problem Statement:

Write a Python program to **swap two numbers** without using a third variable.

Input Format:

Two numbers

Output Format:

Print the swapped values.

11. Convert Decimal to Binary (Using Built-in Function)

Problem Statement:

Write a Python program to convert a given **decimal number to binary** without using loops or conditionals.

Input Format:

Decimal number (int)

Output Format:

Print the binary representation.

☐ *Hint:* Built-in conversion functions.

12. Calculate Absolute Difference

Problem Statement:

Write a Python program to calculate the **absolute difference** between two numbers.

Input Format:

Two integers

Output Format:

Print the absolute difference.